Sandeep Kumar Sahu

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Education

Veer Surendra Sai University of Technology (VSSUT), Burla 2024 – 2028
B.Tech in Computer Science & Engineering (5th Semester) *CGPA: 8.63/10*

SSVM NK Nagar, Berhampur – Class XII (CHSE): 90% | Class X (BSE): 91.16%

Experience

AI & Machine Learning Intern Dec 2025 – Jan 2026
IBM SkillsBuild – Edunet Foundation – AICTE

- Implemented machine learning classification models using Python and Scikit-learn on structured datasets.
- Evaluated model performance using accuracy, F1-score and validation metrics.

Cybersecurity with Generative AI Intern Oct 2025 – Nov 2025
VOIS – Edunet Foundation – AICTE

- Studied cybersecurity threat detection, vulnerability analysis and security monitoring techniques.
- Explored Generative AI applications for cybersecurity analysis and threat awareness.

Projects

AI-Powered Digital Safety Platform | *AWS Student Community Day 2026 – 2nd Prize* *AWS Lambda, API Gateway, ML*

- Developed a cybersecurity platform for phishing detection, spam/message detection, password-breach checking and AI-assisted security guidance.
- Built serverless components using AWS Lambda and API Gateway with integrated machine learning models and security services.

ElderCare+ | *Hack Odisha 2025 – 7th Place* *React, TypeScript, TanStack Query, Leaflet.js*

- Developed a healthcare platform for elderly users with medicine reminders, appointment scheduling and essential support services.
- Integrated map-based functionality to locate nearby pharmacies, hospitals and healthcare resources.

Network Intrusion Detection System *Python, Pandas, NumPy, Scikit-learn*

- Developed an ML-based system to classify malicious network traffic using NSL-KDD and CICIDS2017 datasets with preprocessing and feature engineering.
- Implemented Random Forest and SVM classifiers and evaluated performance using accuracy, precision, recall, F1-score and confusion matrix.

Fake News Detection using Machine Learning *Python, NLP, Scikit-learn, TF-IDF*

- Built an NLP classifier using the WELFake dataset with text preprocessing and TF-IDF vectorization.
- Compared multiple classification algorithms using accuracy, precision, recall and cross-validation.

KnowledgeAI – Retrieval-Augmented Generation System *Python, RAG, LLMs, FAISS, Sentence Transformers*

- Developed a document question-answering pipeline using semantic embeddings, document chunking and vector similarity search.
- Integrated retrieved document context with an LLM to generate context-aware responses and reduce unsupported answers.
- Evaluated retrieval relevance and response quality across representative document queries.

Technical Skills

Languages: Python, JavaScript, TypeScript, SQL
Machine Learning: Supervised & Unsupervised Learning, Feature Engineering, Feature Selection, Hyperparameter Tuning, Cross-Validation, Ensemble Learning, Model Optimization
Deep Learning & NLP: Neural Networks, Transformers, Attention Mechanisms, Text Classification, TF-IDF, Tokenization, Semantic Search, Text Embeddings
Generative AI & RAG: Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Prompt Engineering, Vector Search, Document Chunking, Context Retrieval, LLM Evaluation
ML Libraries: NumPy, Pandas, Scikit-learn, Matplotlib, PyTorch, TensorFlow, Hugging Face Transformers
Web Development: React.js, TypeScript, TanStack Query, Tailwind CSS, Leaflet.js, REST APIs
Cloud & Tools: AWS Lambda, API Gateway, Git, GitHub, Docker, Vercel, Jupyter, Google Colab, VS Code
Systems & Security: Linux, Kali Linux, SSH, Command Line, Network Security, Threat Detection

Leadership Experience

Social Media Team – Enigma, The Web & Coding Club of VSSUT Burla Present

- Manage social-media content and digital promotion for workshops, coding events and technical competitions.

Social Media Team – CSE Society, VSSUT Burla Present

- Create digital content and support online communication for technical events and student programs.

Achievements

- **2nd Prize** – Amazon Code Conquest, AWS Student Community Day 2026.
- **7th Place** – Hack Odisha 2025 Hackathon.
- Reached **Round 5** – Hackiik Cybersecurity Hackathon.
- Participant – Smart India Hackathon 2025 and ISRO National Hackathon 2025.
- Completed OverTheWire Bandit cybersecurity challenges (Levels 0–24).

Certifications

AI & ML Fundamentals – IBM: Machine learning fundamentals, data preprocessing, supervised learning, model training and evaluation.
Cybersecurity with Generative AI – VOIS: Cybersecurity fundamentals, threat detection, vulnerability analysis and applications of Generative AI in security.
Frontend Web Development – IBM/AICTE: HTML, CSS, JavaScript, responsive web design and fundamentals of modern frontend development.
Linux Command Line & Cybersecurity – OverTheWire: Linux command-line operations, file permissions, shell utilities and practical cybersecurity challenges through Bandit Levels 0–24.